Towards Understanding the Energy Consumption of Virtual Reality Applications in Gaming, Education, and Entertainment

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ABSTRACT

Context. Virtual reality (VR) is on the rise in popularity, yet its notable energy consumption demands attention. With the expanding adoption of VR, it becomes crucial to comprehend the energy consumption trends among various VR applications and explore strategies for minimizing their energy usage.
Goal. We aim to empirically assess the energy consumption in widely-used VR applications. Our focus is on (i) identifying the influence of energy-intensive components within VR towards the overall energy consumption of VR applications and (ii) comparing the differences in these factors across popular VR applications.
Method. We use a Meta Quest 2 headset to assess energy-intensive components in 9 VR applications across 3 different domains. Adopting a 'multiple factors - single treatment' (MF-ST) design, we collect energy measures and statistically analyze them.
Results. The study provides evidence that display, storage, memory, CPU, and GPU utilization correlate weakly with the energy consumption of VR applications. It also emerged that the energy consumption of VR applications may be influenced by factors related to usage scenarios, but less so by the specific applications used. The energy consumption of VR games is significantly lower than that of Entertainment and Education VR applications.
Conclusions. This is a first step towards creating awareness and providing evidence-based results to VR developers for designing and developing greener VR applications.

ACM Reference Format:
Jyothsna Priya Kattakinda, Sachin Dhananjaya, Samuel Mojžiš, Amine Nadif, Mattheus Hanna, Radu Apsan, Ivano Malavolta. 2024. Towards Understanding the Energy Consumption of Virtual Reality Applications in Gaming, Education, and Entertainment. In *2024 ACM/IEEE 8th International Workshop on Games and Software Engineering (GAS '24), April 14, 2024, Lisbon, Portugal.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3643658.3643922

1 INTRODUCTION

The concept of metaverse is based on the vision of having a vast evolving virtual realm blending with the physical reality. Although the term was coined in the year 1992 [1], it was not until 2003 that the first meta platform "Second life" came into the picture.

Virtual Reality (VR) lies at the core of the metaverse, serving as its foundational technology. VR employs 3D computer modeling – a captivating form of graphic design – to immerse users within 3D virtual environments [2]. Among the many VR companies, a prominent one is Meta. Meta's fundamental mission is to empower individuals by granting them the ability to create communities and foster closer connections globally. Meta's journey commenced with the transformative impact of Facebook in 2004, reshaping global connections. Over the years, platforms like Messenger, Instagram, and WhatsApp have amplified this transformative power. Meta now endeavours to transcend the constraints of traditional 2D screens, popularizing immersive experiences in the metaverse. Their main products involve VR headsets and smart glasses. Among them is the Meta Quest 2, which is the VR headset we use in this study.

VR headsets use a diverse set of components, such as projection units, proximity sensors, eye tracking, and powerful processing units. Understanding the energy consumption patterns amongst the components in these systems is vital for fostering sustainable technology development. Given the growing concerns about the environmental impact of high energy usage, researching the energy footprint of these technologies is crucial. By identifying the components driving energy consumption and optimizing their design, we can move towards a greener approach, utilizing energy-efficient technologies and promoting the integration of sustainable practices.

The **goal** of this study is to empirically assess the influence of energy-intensive components towards the overall energy consumption of VR applications. Specifically, we use a Meta Quest 2 headset to assess energy-intensive components in 9 VR applications across 3 different domains. In our statistical analysis of the collected measures, we found differences in energy consumption across different VR domains. The study identifies the most energy-intensive components and provides evidence that the energy consumption in VR is influenced by factors related to usage scenarios, but less so by the specific applications used.

The **main contributions** of this study are (i) an empirical assessment of the energy consumption of 9 VR applications across the Gaming, Education, and Entertainment domains, (ii) a discussion about obtained results and future research directions, and (iii) the full replication package of the study containing raw data, source code, and scripts for data analysis. This experiment aims to provide insights into where the energy is predominantly consumed by VR applications and potentially guide sustainable design or optimization efforts in the future.

This study has two **main implications for VR developers**. First of all, the usage of hardware components like the display, storage, memory, CPU, and GPU tends to correlate with energy consumption, but only weakly; this might be an indication that

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GAS '24 , April 14, 2024, Lisbon, Portugal
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ACM ISBN 979-8-4007-0561-8/24/04…$15.00
https://doi.org/10.1145/3643658.3643922

energy consumption is influenced by other factors, such as the specific choices and frameworks used by VR developers at the software level; further research is needed to confirm this claim, but we suggest VR developers to follow green software engineering best practices like those presented in [3, 4] in their VR applications. Second, Gaming VR applications tend to significantly consume less energy than other VR domains (*i.e.,* Education and Entertainment, in this experiment); we speculate that VR engines specific to gaming are optimized for higher performance and usage of resources, which in turn might lead to lower energy consumption.

2 RELATED WORK

Jansen *et al.* [5] evaluated the performance of Meta's flagship virtual reality device, the Quest Pro, in the context of the metaverse. Their study highlighted the battery's faster drain when using wireless offloading and the performance of outdated WiFi4. Our study analyzes the energy distribution across the Meta Quest 2's hardware components, like the display and memory, to identify energy consumption, in contrast to their focus on battery life.

The study by Huang *et al.* [6] delves into the energy inefficiencies associated with current VR systems, particularly in the context of spherical panoramic video processing. They identify that techniques designed for conventional planar videos are being repurposed for VR, leading to increased energy consumption. Whereas, their research focuses on optimizing VR video processing for energy efficiency, our study specifically examines the Meta Quest 2's energy distribution during various VR gameplay sessions.

The study by Chuah *et al.* [7] explores cloud gaming as a sustainable solution for advanced video gaming, which typically demands high computational power for realistic graphics rendering. While their research delves into the broader realm of cloud gaming and its green implications, our study is concentrated on the Meta Quest 2, focusing on its energy distribution patterns during VR gameplay.

The study by Webb *et al.* [8] delves into the energy consumption of high-definition game consoles, highlighting the rising energy use due to increased ownership, use, and enhanced performance. The study offers a detailed breakdown of energy use across different usage modes. Contrasting with their focus on high-definition game consoles, our research is specifically tailored to the Meta Quest 2, analyzing its energy distribution during VR gameplay sessions.

The research by Mills *et al.* [9] investigates the energy consumption of gaming computers, noting that gaming is the most energy-intensive use of PCs. With gaming systems being the fastest-growing platform, the study highlights vast variations in power ratings for different computer components, indicating substantial energy-saving potential. In contrast to their broad exploration of gaming computer energy consumption, our study is centered on the Meta Quest 2, examining its specific energy distribution during VR gameplay.

Another study by Mills *et al.* [10] delves into the energy consumption of computer gaming, measuring 26 different systems and highlighting the variance based on hardware and game titles. It underscores that U.S. gaming's energy expenditure stands at 5 billion annually. While their focus spans various gaming platforms, our research specifically targets the energy patterns of the Meta Quest 2 during VR gameplay.

The metaverse's environmental implications are explored by researchers who surveyed Romanian IT professionals [11], revealing concerns like increased power consumption and CO2 emissions within the metaverse. Their findings emphasize the need for international regulations and sustainable policies to mitigate these impacts. Unlike their broader focus on the metaverse, our research narrows in on the Meta Quest 2's energy consumption patterns during VR gaming sessions.

3 EXPERIMENT DEFINITION

3.1 Goal, Research Questions, and Metrics

The overarching goal of this experiment is to analyse the influence of main components in VR devices towards the overall energy consumption and compare the differences in these factors across the most popular VR applications. The above-mentioned goal is achieved by answering the following research questions.

RQ1: How do the energy-intensive components within VR device, influence overall energy consumption?

Approach: VR devices can be classified into two categories which can be tethered to a base computing device and standalone untethered devices that has built-in computing device [12]. Our experiment uses the tethered device Meta Quest 2. This can be identified by calculating the power consumption in the Meta Quest 2. We then compare how the total power consumption is influenced by the energy-intensive components. We achieve this by initially identifying the energy-intensive components. Then, a correlation can be derived to calculate the energy consumption.

RQ2: What is the energy consumption profile of different VR application domains and experiences?

Approach: From findings obtained through RQ1, we calculate the energy consumption using the power consumption of different VR application domains and identify if there are any significant differences in energy usage by performing an in-depth comparison between them.

For RQ1, the power consumption for the device can be attributed to 4 major components *i.e.,* display, storage (eMMC), memory (DRAM), and compute (SoC) [12]. We initially compare the influence of the 4 components towards the power consumption. And then identify the correlation between the power consumption and energy distribution.

Below are the metrics used to calculate the Total Power consumption:

(1) **Voltage (in Volts (V)):** Voltage indicates the *pressure* that pushes electricity. The amount of power voltage is indicated by volt (V), and higher voltages cause more electricity to flow to an electronic device [13].
(2) **Current (in Amperes (A)):** Current is the rate at which electrons flow past a point in a complete electrical circuit [14].

The Power is calculated using the formula,

$$Power\ (P)\ =\ Voltage\ (V)\quad X\quad Current\ (C) \tag{1}$$

Where, Power (P) is measured in Watts (W), Voltage (V) is measured in Volts (V) and Current (C) is measured in Amperes (A).

The rationale behind choosing these 4 components related to hardware is that energy is not consumed by the software directly, but rather that its functioning involves hardware which consumes energy [15].

Below is the detailed summary of the chosen components:

(1) **Display:** Display plays an important role in the performance of the VR devices [16]. Meta Quest 2 uses an LCD display.
(2) **Storage:** In the VR device, in order to perform operations, the application uses the embedded MultiMediaCard (eMMC) storage within the device while in use [17]. Meta Quest 2 has a storage of 128GB.
(3) **Memory:** This component indicates the Dynamic Random-Access Memory (DRAM) used embedded in VR device used by the application while performing its operations [12].
(4) **Compute (SoC):** The Meta Quest 2 uses the Snapdragon XR2, a system on a chip (SoC) by Qualcomm. The SoC handles graphics, video and display processing support of 4K videos [18].

We calculate energy consumption using the formula below [19]:

$$Energy\ (E)\ =\ Power\ (P)\quad X\quad Time\ (t) \tag{2}$$

Where, Energy is measured in Joules (J), Power (P) is measured in watts (W) and Time (t) is the elapsed time [19]. We compare the power consumption and the components to determine if the power consumption is influenced by the energy-intensive components.

For RQ2, the research will be performed on the most popular VR application domains: Gaming, Entertainment, and Education [20]. Based on the results generated from addressing RQ1, we perform a comparison of these metrics across VR application domains.

4 EXPERIMENT PLANNING

This research aims to compare the energy consumption profile of different VR applications and experiences by calculating the power consumption. The complete replication package of this study is publicly available[1] to facilitate independent replication and verification of the obtained results.

4.1 Subjects Selection

Given the extensive diversity in the VR ecosystem, we narrow down our subjects to ensure a focused and manageable study. The population of distinct VR devices is extensive, with numerous models and specifications. In this study we use the Meta Quest 2 since it is one of the most used VR headsets in the market; for example, as of September 2023, the Meta Quest 2 has about 41.1% of the market share on Steam, one of the leading video game digital distribution services [21].

VR applications and experiences are available in abundance across various platforms. To simplify our experiment, we select the VR domains: *gaming, education and entertainment* based on the popularity of those VR domains [20].

Another important decision is the VR applications to be tested. For this, we chose the top-rated VR applications[2] across each VR domain. In order to run the test runs without any inconvenience we ensured all these applications are available for free on the Meta Quest 2. The other important criterion that was also taken into consideration was whether these applications need internet connectivity. Many top-rated applications supported by Meta[3] function without the need for internet connection. To maintain the accuracy and correctness of the experiment we ensured to choose the applications that require internet connection to ensure accurate results. After filtering the applications based on the above-mentioned criteria (refer Table 1) we focus our experiment on 9 VR applications in total, where for each of the three VR domains, the three top-rated applications are selected. This can be found in Table 2. In selecting the VR applications for our study, we acknowledge that the chosen application may not be completely representative of the entire domain under investigation. Recognizing the diversity inherent in the domain, we acknowledge that variations among VR applications can arise from differences in functionalities, user interfaces, and underlying technologies. The primary rationale for subject selection for this experiment is to study the energy consumption of different VR applications that are popular within the VR community. We chose popularity as the key factor as it ensures that our research findings are applicable to a significant user base, providing insights that align with real-world VR usage patterns.

Table 1: Inclusion and exclusion criteria for subjects

Criteria	Description
Inclusion	Popular VR domains
Inclusion	Top ranked VR applications
Exclusion	Paid VR applications
Exclusion	VR applications that do not require Internet Connection.

Table 2: Subjects of this study

VR Domain	VR Application	Rating (# of ratings)
Gaming	Gym Class - Basketball VR (A1)	4.87 (23,487)
Gaming	Gorilla Tag (A2)	4.69 (51,911)
Gaming	Blaston (A3)	4.52 (2,298)
Education	Innerworld (A4)	4.73 (338)
Education	ShapesXR (A5)	4.24 (193)
Education	Zoe(A6)	3.6 (27)
Entertainment	RecRoom (A7)	4.63 (1,370)
Entertainment	VR Animation Player (A8)	4.27 (409)
Entertainment	Ultimechs (A9)	4.18(3546)

4.2 Experimental Variables

Based on the two Research questions, we infer nine independent variables and seven confounding variables for our experiment.For RQ1 the subjects are the nine applications we chose: *Gym Class - Basketball VR (A1), Gorilla Tag (A2), Blaston (A3), Innerworld (A4), ShapesXR (A5), Zoe (A6), RecRoom (A7), VR Animation Player (A8), Ultimechs (A9).* For a detailed description of the applications, see Section 4.1. The independent variables are derived from experiment metrics: *Voltage (V), Current (C)* and components: *Display (D), Storage (S), Memory (M) and Compute (C).* The dependent variable for RQ1 is the *Total Power Consumption (TP)* for different VR applications chosen from the popular VR domains: Gym Class - Basketball VR (TP1), Gorilla Tag (TP2), Blaston (TP3), Innerworld (TP4), Supernatural (TP5), ShapesXR (TP6), Oculus First Contact (TP7), VR Animation Player (TP8) and Epic Roller Coasters (TP9).

[1] https://github.com/S2-group/gas-2024-energy-vr-apps-rep-pkg
[2] https://www.roadtovr.com/best-oculus-quest-2-games-apps-rated-january-2023/3/
[3] https://www.meta.com/en-gb/experiences/view/777073612853145/

The dependent variable *Total Power Consumption (TP)* is the multiplication of confounding variables Voltage (V) and Current (C). This dependent variable is calculated using the formula,

$$TP_i = V_i \quad x \quad C_i \tag{3}$$

where $i \in \{A1, A2, A3, A4, A5, A6, A7, A8, A9\}$.

For RQ2 the dependent variables are the energy consumption for different VR applications chosen from the popular VR domains: *Gym Class - Basketball VR (EC1), Gorilla Tag (EC2), Blaston (EC3), Innerworld (EC4), Supernatural (EC5), ShapesXR (EC6), Oculus First Contact (EC7), VR Animation Player (EC8) and Epic Roller Coasters (EC9).* For the detailed description for the chosen applications, see Section 4.1. Energy Consumption (EC) is calculated as follows:

$$EC_i = TP_i \quad X \quad T\ (t) \tag{4}$$

where $i \in \{A1, A2, A3, A4, A5, A6, A7, A8, A9\}$ and T is time. Here, Time (t) is constant as all the iterations are run for 30 min, see Section 4.4 for more details.

4.3 Experimental Hypotheses

In this section, we define and present the experimental hypotheses for the research questions posed in this study. The hypotheses are structured to investigate the energy consumption patterns within VR devices, focusing on energy-intensive components and their variations across popular VR application domains.

For RQ1, we aim to determine how the energy-intensive components within VR devices influence the overall energy consumption. The independent variables represent the current and voltage of the Meta Quest 2 VR device. The dependent variable is the total power consumption (*TP*) for a variety of selected VR applications.

To answer RQ1, we have constructed the following hypothesis:

$$H_0^{\text{PC}} : \mu_D = \mu_S = \mu_M = \mu_C \tag{5}$$

Here, the independent variables showcase mean value of energy-intensive components (*Display (D), Storage (SP), Memory (P), Compute (C)*). Dependent Variable: Total Power Consumption (*TP*) for different VR applications (TP_1 *to* TP_9). The null hypothesis states that there is no significant difference in the mean values of the identified energy-intensive components (D, S, M, C) for the selected VR applications. This leads to the alternative hypothesis which states that there is a significantly relevant difference in the energy-intensive components (*D, S, M, C*) for for at least one of the selected applications.

$$H_1^{\text{PC}} : \text{At least one among } \mu_i \text{ is different} \tag{6}$$

where $i \in \{D, S, M, C\}$.

For RQ2, the aim is to analyze the energy consumption of each VR application within the specified constant time frame, allowing for a fair comparison of the energy profiles across the chosen VR application domains. The hypotheses are structured to assess if there are significant differences in energy consumption among these domains, enabling a comprehensive understanding of the VR components with the highest energy consumption.

$$H_0^{\text{EC}} : \mu_{EC1} = \mu_{EC2} = \mu_{EC3} \tag{7}$$

where μ_{EC1} to μ_{EC3} define the mean energy consumption within each domain. The null hypothesis suggests that the mean energy consumption (μ_{EC}) for all selected VR application domains is equal. In other words, there is no significant difference in energy consumption across these application domains. On the contrary, the alternative hypothesis suggests that at least one mean energy consumption (μ_{EC}) is different, indicating that there might be variations in energy usage among the VR application domains.

$$H_1^{\text{EC}} : \text{At least one among } \mu_{EC} \text{ is different} \tag{8}$$

By examining and comparing the energy consumption of VR applications within each domain, we aim to identify potential differences in their energy profiles, contributing to a comprehensive understanding of energy usage patterns in different VR experiences.

4.4 Experiment Design

The design type we have chosen is "multiple factors - single treatment" design type (MF-ST). The treatments we consider correspond to the factors. These factors are associated with the metrics used to measure the power and energy consumption of the selected VR applications. We have selected VR applications based on their popularity ranking as outlined in Table 2. For the experiment, we have decided to pick 3 applications from 3 categories (see Table 2), which makes it 9 applications in total to get data from during the run time.

To address RQ2, it is essential to ensure consistent run time for each iteration when running the application. The VR applications can vary in complexity and specifications. To ensure consistency in their run times for effective data comparison, we set the time as constant for all the applications. We have selected a specific time allocation: *30 minutes* for each application run. This duration is practical and manageable, and strikes a balance between obtaining detailed insights and minimizing the overall duration of the experiment or potential external disturbances that could compromise the accuracy of our measurements. Through our analysis of these selected applications, we aim to identify the specific components of VR that consume the most energy. Therefore, each metric is measured on each application, and each application will be executed three times, which is enough to account for potential variability or outliers in the data. It provides more reliable average measurements and allows us to assess the consistency of results across multiple runs. This approach will provide us with the insights needed to determine which VR components have the highest energy consumption, thereby effectively addressing RQ2.

To calculate the execution time of the experiment, we consider the number of applications, repetitions and run time of each application. The formula then looks like this: 9 applications x 3 repetitions x 30 minutes. This results in a total of 27 test runs, spanning a duration of 810 minutes (equivalent to 13.5 hours) making us able to achieve the methodological rigour in our experiment.

Our objective is to collect data for every metric within a single run. During each run, we specifically measure the defined metrics related to power consumption in watts, and each application for the energy consumption in joules. Based on the gathered data we will then perform an analysis with the goal to answer our research questions.

4.5 Data Analysis

4.5.1 Data Exploration. In the data exploration, we perform two main tasks, (i) *understanding the data* and (ii) *normality checks*. In

this first step, we explore the collected data and analyse it using descriptive statistics. In order to select appropriate statistical tests it is essential to estimate if the collected data is normally distributed or not. Normality checks can be performed using statistical tests like the Shaprio-Wilk test, Kolmogorov–Smirnov test, or visual inspection using box plots, P–P plots, Q-Q plots or histograms [22]. For our experiment, we have used the Shaprio-Wilk test and Q-Q plots to test the normality of the data. The Shaprio-Wilk test is more appropriate for small sample data. For the Shaprio-Wilk test, the null hypothesis states that data is considered from a normally distributed population. When $P > 0.05$, the null hypothesis is accepted and data is said to be normally distributed. Whereas, the Q–Q plot is a scatterplot generated by plotting two sets of quantiles (observed and expected) against one another. For normally distributed data, observed data are approximate to the expected data [22].

4.5.2 Inferential Analysis . In inferential analysis, we perform two main tasks, *hypothesis testing and effect size estimation.* Hypothesis Testing is used to determine whether the given data satisfies the hypothetical condition. Generally, the statistical tests are either parametric or non-parametric. In parametric tests, it is assumed that the data follows a specific distribution, often the normal distribution. Whereas, non-parametric tests make fewer distributional assumptions [23]. So in our experiment, we perform either a parametric test or a non-parametric test based on the results obtained from the previous stage. For RQ1, we perform hypothesis testing with correlation and choose the statistical test based on whether the data is normally distributed or not.

Normality assumed: Pearson's Correlation Coefficient (r) describes the strength and direction of the linear relationship between two quantitative variables [24]. We perform this test to validate the hypothesis of RQ1. For RQ2, we perform an ANOVA test. In this test, the variance within each group defined by the independent variable is first examined. Then, the variance within each group is compared with the overall variance of the group means. A large difference in means combined with small variances within the groups signifies a greater difference between the groups. In this case, the null hypothesis is rejected.

Normality not assumed: Spearman's Rank-Order Correlation hypothesis testing is performed in order to determine if the hypothesis for RQ1 is valid. Spearman's correlation determines the strength and direction of the monotonic relationship between two variables. It helps to determine if a variable value increases or decreases based on another variable [25]. For RQ2, we perform the Kruskal-Wallis Test or one-way ANOVA on ranks. The test determines whether the medians of two or more groups are different. However, this test only determines if there is a significant difference between groups. So, in addition, we perform effect size estimation to determine which groups are different [26].

Effect size estimation: It is a statistical concept that quantifies the magnitude or strength of a relationship or difference between groups. It provides a measure of how practically significant an observed effect or relationship is. For this, we use Cliff's Delta for pairwise comparisons between the domains to determine the magnitude of the differences in energy consumption. Cliff's Delta measures the probability that a randomly selected data point from one group is greater than a randomly selected data point from another group.

5 EXPERIMENT EXECUTION

The experiment is executed using a systematic approach to maintain the integrity and precision of the results. The values for the dependent variables of the experiment are collected via the OVR Metrics Tool, a specialized tool developed by Meta for the Meta Quest platform[4]. The OVR Metrics tool is designed to provide detailed performance information about running applications, offering insights similar to those provided by VrApi Logcat logs but accessible from an on-device application. The tool operates in two modes: Report Mode and Performance HUD Mode. In Report Mode, it records performance reports about a VR session, which can be exported as a CSV file along with PNG images after the session has ended. On the other hand, the performance HUD mode displays a real-time HUD overlay on running applications, presenting instantaneous performance graphs and information. The information displayed can be customized according to your preferences. The tool is instrumental in collecting accurate and reliable data on the performance and energy consumption of Meta Quest 2 during VR application sessions [27]. The OVR Metrics Tool is carefully configured to record crucial metrics essential for evaluating energy consumption, such as TimeWarp GPU Time for display, GPU and CPU utilization for compute, used memory for storage, Application VSS, RSS, and Dalvik PSS for memory. It is important to mention that the Meta Quest 2 turns off if the device is not worn by the user, which is why this OVR Metrics Tool is turned on for recording just before using the application, and stopping the recording right after the application is played for 30 minutes to ensure precise data is obtained. The detailed description of the OVR statistics can be found in the Table 3. Following each session, the collected data is exported to a CSV file via the OVR Metrics Tool. A stringent data cleansing process is performed to filter out any discrepancies, noise, or irrelevant data to uphold the integrity of the data set. The metrics provided in Table 3 are extracted from each data file. The Power Voltage and Power Current values are first converted to Volts and Amperes respectively. These values are then used to calculate the Power Consumption. The other metrics such as display, storage, memory, and compute are individually used to identify the correlation between these metrics and the total power consumption. Each application runs for 30 minutes, undergoing multiple iterations (3x) to enhance the reliability of the data.

To facilitate the execution of the experiment and the extraction of data, Bash is used to execute commands through the Android Debug Bridge (ADB), providing a streamlined interface for interacting with the device and managing the data efficiently. Android Debug Bridge (ADB) [28] is a command-line tool essential for our experiment, allowing communication and command execution on the Meta Quest 2 device.

For the execution of the experiment, Meta Quest 2 with 128GB storage serves as the primary hardware. The Meta Quest 2, successor of the original Oculus Quest uses the Snapdragon XR2, a

[4]https://developer.oculus.com/downloads/package/ovr-metrics-tool/

Table 3: OVR Metrics

Name	OVR Metric	Description
Display	TimeWarp GPU Time (timewarp_gpu_time _microseconds)	Represents the amount of time ATW (Asynchronous Time Rawp) takes to render. Correlates to the number of layers used and their complexity.
Storage	Used Memory (app_pss_MB)	Amount of memory storage utilized by an application during runtime.
Memory	Application VSS, RSS, Dalvik PSS (app_vss_MB)	Represents the various memory metrics, indicates memory usage of application.
Compute	GPU and CPU Utilization	Represents the utilization of the device's processing units.
Current	Power Current (power_current)	Shows the power current of the headset in milliamps.
Voltage	Power Voltage (power_voltage)	Indicates the power voltage in millivolts.

Qualcomm chip-based system, designed explicitly for VR and augmented reality devices, coupled with 6GB of LPDDR4X RAM. The device features a single, fast-switch LCD panel with a per-eye resolution of 1832×1920 and a refresh rate of up to 120 Hz. The Meta Quest 2 operates on Android, requiring a smartphone running the Meta Quest app for the initial setup. It supports multiple accounts with the ability to share purchased software between them and has received several updates, enhancing its features, such as adding *Air Link* for streaming games from a PC over Wi-Fi and experimental APIs for augmented reality features [29].

6 RESULTS

6.1 Data exploration

6.1.1 Understanding the data. In order to understand the data recorded by OVR Metrics Tool, we performed a descriptive statistical analysis of the data. The recorded data contains Voltage and Current in terms of milli-volts and milli-amperes respectively. Data transformation has been performed to convert these variables into numerical values of units of volts and amperes respectively. The power consumption is calculated using these two variables, the descriptive statistics of these confounding and dependent variables can be found in Tables 4, 5, and 6.

Table 4: Descriptive Statistics for the Gaming Domain

Applications	Gym Class - Basketball VR (A1)			Gorilla Tag (A2)			Blaston (A3)		
Variables	PC	PV	TP	PC	PV	TP	PC	PV	TP
Min	0.89	3.613	3.51	0.889	3.56	3.74	0.965	3.457	3.65
1st Quantile	1.5	3.7	5.49	1.5	3.61	5.42	1.5	3.61	5.38
Median / 2nd Quantile	1.5	3.75	5.6	1.5	3.66	5.47	1.5	4.013	5.95
3rd Quantile	1.5	3.94	5.79	1.5	3.98	5.94	1.5	4.1	6.09
Mean	1.47	3.80	5.58	1.49	3.77	5.62	1.48	3.911	5.78
Max	1.5	4.086	6.11	1.5	4.215	6.32	1.5	4.239	6.31
St. Deviation	0.095	0.139	0.382	0.049	0.197	0.317	0.054	0.239	0.371
Count	4549	4549	4549	5508	5508	5508	5256	5256	5256

Table 5: Descriptive Statistics for the Education Domain

Applications	Innerworld (A4)			ShapesXR (A5)			Zoe (A6)		
Variables	PC	PV	TP	PC	PV	TP	PC	PV	TP
Min	0.842	3.512	3.47	0.83	3.702	3.52	1.073	3.457	4.24
1st Quantile	1.44	3.65	5.38	1.48	3.77	5.59	1.5	3.57	5.35
Median / 2nd Quantile	1.5	3.77	5.69	1.5	3.819	5.68	1.5	3.8225	5.73
3rd Quantile	1.5	3.98	5.79	1.5	4	5.97	1.5	4.03	6.01
Mean	1.45	3.81	5.538	1.48	3.877	5.744	1.49	3.822	5.70
Max	1.5	4.212	6.32	1.5	4.247	6.34	1.5	4.247	6.35
St. Deviation	0.079	0.19	0.262	0.047	0.138	0.257	0.034	0.226	0.339
Count	6149	6149	6149	5366	5366	5366	5290	5290	5290

Table 6: Descriptive Statistics for the Entertainment Domain

Applications	RecRoom (A7)			VR Animation Player (A8)			Ultimechs (A9)		
Variables	PC	PV	TP	PC	PV	TP	PC	PV	TP
Min	0.966	3.59	3.67	0.84	3.572	3.38	0.931	3.533	3.57
1st Quantile	1.48	3.64	5.41	1.06	3.65	4.2	1.5	3.63	5.42
Median / 2nd Quantile	1.5	3.669	5.47	1.142	4.073	4.475	1.5	3.793	5.67
3rd Quantile	1.5	3.7	5.51	1.5	4.14	4.85	1.5	3.87	5.8
Mean	1.47	3.67	5.405	1.15	3.95	4.54	1.48	3.765	5.600
Max	1.5	3.808	5.67	1.5	4.204	6.25	1.5	4.029	6.04
St. Deviation	0.060	0.037	0.210	0.127	0.230	0.501	0.060	0.133	0.283
Count	5212	5212	5212	5008	5008	5008	5389	5389	5389

6.1.2 Normality checks. We used Q-Q plots and the Shapiro-Wilk test to check if the distribution of variables follows a normal distribution. For the sake of space, Q-Q plots are not reported here, but are included in its replication package. The Q-Q plots indicate that our collected data does not follow a normal distribution.

On applying the Shapiro test to the energy consumption across different applications, we see that the p-value for each application is much less than 0.05. Thereby, we can say that the energy consumption does not follow a normal distribution.

6.2 Statistical Analysis

6.2.1 Hypothesis Testing. Before testing our hypothesis, we perform a statistical test to measure if the subjects consume different amounts of energy. For this, we use one way-ANOVA test. The F-statistic of the ANOVA test is 2.0140, with a p-value of 0.2142. The non-significant p-value indicates that there is insufficient evidence to reject the null hypothesis. While the analysis does not reveal significant differences, it perhaps suggests that other factors such as energy-intensive components, not considered in this specific analysis, may contribute to variations in power consumption. So for this, we perform further tests to prove our hypothesis. Since our data is not normally distributed, in order to test our hypothesis, we decided to go with Spearman's Rank-order correlation for RQ1 and the Kruskal-Wallis test for RQ2.

Research question 1 – To identify the correlation between energy consumption and the energy-intensive components, all the data has been combined for all 9 applications to get close to accurate results. Figure 1 depicts the monotonic relationship between Power and different energy-intensive components. For comprehensible visualization, this graph is generated to depict the relationship for randomly chosen 5000 data samples. Table 7 contains Spearman's correlation coefficient r, which is calculated for total data *i.e.,,* 47727 records. While the correlations are generally positive, which means that as the confounding variables (*display, storage, memory, CPU utilization, GPU utilization*) increase, power consumption tends to increase. This aligns with our hypothesis 5, that these components influence power consumption. But all the correlation coefficients range from 0.19 to 0.34, indicating very weak to weak relationships [30]. This suggests that relationships between these variables and power consumption are not very robust and that the overall power consumption could be affected by other factors in the VR device.

Table 7: Results of the correlation analysis

	r-Value	Relationship
Display - Power	0.34	Positive - weak
Storage - Power	0.19	Positive - very weak
Memory - Power	0.24	Positive - weak
CPU Utilization - Power	0.20	Positive - weak
GPU Utilization - Power	0.34	Positive - weak

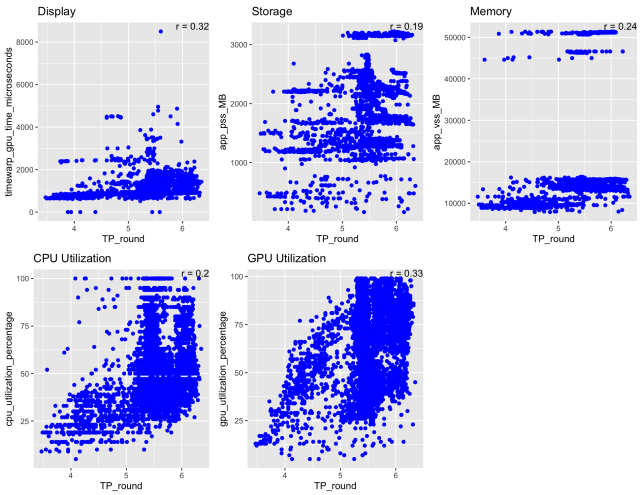


Figure 1: Spearman's Rank-Order Correlation

Research question 2 – For Hypothesis 7, on testing the energy consumption against domain type and application, the p-values of the Kruskal-Wallis test are both below the significance level of 0.05. This suggests that the differences in energy consumption across the different domains and across different applications are highly significant. But this does not tell us enough about the difference in energy consumption across different domains.

We estimate the difference between the domains via the Cliff's delta coefficient. As shown in Table 8, the value for Education and Entertainment suggests that there is a small difference in the energy consumption between the two of them, which suggests it is not practically relevant. On the other hand, by observing the other two negative values in the table for Entertainment and Education, we can observe that gaming has much lower values of energy consumption when compared to Entertainment and Education domains.

Table 8: Effect size estimates between Domains

Domain 1	Domain 2	Delta estimate
Education	Entertainment	0.071
Entertainment	Gaming	-0.3773
Education	Gaming	-0.3681

7 DISCUSSION

Reflections on obtained results – Our objective when targeting RQ1 was to find the influence of energy-intensive components on the overall energy consumption of a VR application. With the results of the Spearman's Rank-order correlation, it is clear that energy-intensive components play a role, but it is also evident that other factors contribute to power consumption as the relationship resulted from very weak to weak. Moreover, our data set contained 47727 records, and correlation analysis best results are yielded when the sample size is large. This would help in getting precise results and potentially stronger relationships. The results also emphasize the importance of a cautious interpretation of the obtained results. A correlation between variables does not imply causation, and it is important to consider the potential influence of confounding variables that were not part of this study.

Our objective when targeting RQ2 was to check energy consumption across different domains. Contrary to our initial expectations, the QQ plots and the Shapiro-Wilks test revealed that both energy consumption across different applications and power consumption for various applications did not adhere to a normal distribution. Since the data is not normally distributed, we decided to test our hypothesis using a non-parametric statistical test. The resulting p-values for energy consumption concerning both the domain and the application were significantly less than 0.05. This led us to reject the null hypothesis, indicating a strong difference in energy consumption across various domains and applications. While our findings strongly support the existence of a significant difference in energy consumption, we delved deeper into the nuances of these variations by employing the Cliff's Delta metric. Interestingly, we observed that the Education and Entertainment domains exhibited relatively high energy consumption values, with minimal discernible differences between them. However, when compared to Gaming, both the other domains seem to have higher values outpacing it in terms of energy consumption. In conclusion, while the Kruskal-Wallis test provided robust evidence for rejecting the null hypothesis, the analysis with Cliff's Delta highlighted a significant difference only between Gaming and the other two domains. Our conjecture is that VR engines specific to gaming, such as Unreal, are optimized for higher performance and usage of resources, which in turn might lead to lower energy consumption. Further exploration into domain-specific applications and in-depth investigation may offer additional clarity and a definitive answer to the underlying hypothesis.

Future research directions – We envision the following future research directions toward a better understanding of the energy consumption of VR applications. First of all, the general validity of the results obtained in this study must be further expanded and confirmed. For example, it is important to examine a broader range of VR devices across several manufacturers, including both paired and standalone headsets, and involving a much larger set of VR applications. Moreover, it is relevant to investigate how VR applications manage the interactions and movements of VR users (*e.g.,* how the software reacts to controller, movements, and hand gesture events) and their impact on the energy consumption of the VR headset. Also, VR games and applications usually come with VR-specific settings (*e.g.,* graphics quality, frames prerendering, variable refresh rate); it can be valuable to investigate the impact of such different VR settings on the energy consumption of VR applications. These extensions, when combined, would contribute to a more holistic understanding of energy consumption in the software running in VR applications and its practical implications for VR software developers.

8 THREATS TO VALIDITY

Internal Validity – Our study's internal validity may be influenced by factors like the Meta Quest 2's battery life consistency and ambient temperature fluctuations. These elements could lead to variations in energy consumption measurements. Despite efforts to standardize experimental conditions, the dynamic nature of VR technology and its interaction with these variables could impact the consistency and reliability of our findings. Moreover, the repeatability of movements in different VR applications presents an additional

layer of complexity. The variance in user movements across diverse applications, ranging from minimal to intensive, could significantly affect energy consumption patterns, a factor not comprehensively addressed in our initial setup.

External Validity – The energy consumption trends observed may not accurately reflect the performance of different VR devices or applications not covered in this study. Additionally, the experimental setting, while controlled, might not entirely mimic real-world usage conditions, potentially limiting the generalizability of our results. The influence of user movement, which can vary greatly in everyday use compared to a controlled environment, remains an underexplored aspect that could impact the wider applicability of our conclusions. conclusions.

Construct Validity – Our reliance on the OVR Metrics Tool for energy consumption measurements, while robust, might not fully capture the spectrum of activities that drain energy in VR applications. The tool's granularity and the specific metrics we focused on may overlook aspects of energy consumption, particularly related to user movement and interaction within VR environments. This oversight might lead to an incomplete understanding of how various types of movements, whether significant or minimal, contribute to the overall energy profile of VR applications.

Conclusion Validity – Interpreting data from complex systems like VR requires careful consideration. The multifaceted nature of VR technology, combined with the various factors that contribute to energy consumption, introduces a level of uncertainty in our conclusions. While our analysis is thorough, it is important to recognize the potential for error margins and the impact of unconsidered variables, such as the energy implications of different movement patterns in VR applications. These unaccounted factors necessitate a cautious approach when drawing conclusions from our study. Also, we are aware that each measured VR application has been executed only for 90 minutes, potentially leading to unreliable results; nevertheless, the standard deviation of the collected measures for each VR app tends to be low, making us reasonably confident about the reliability of the obtained results. In any case, as future work, we are planning to repeat the execution of each measured VR app at least 10 times to account for this potential threat to validity.

9 CONCLUSIONS

The study examined energy consumption during virtual reality (VR) experiences using the Meta Quest 2 headset and popular VR applications. The analysis showed that energy consumption patterns were not normally distributed (possibly due to sample size limitations). The Kruskal-Wallis test indicated significant differences in energy consumption between different VR domains and between different VR applications. This suggests that energy consumption in VR may be influenced by factors related to usage scenarios and type of applications.

Acknowledgements. This project is partially supported by the European COST programme under action identifier CA19135 (CERCIRAS).

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